

SEMESTER-IV

URA301: SMART MATERIALS FOR ROBOTICS				
	L	T	P	Cr
	2	0	2	3.0
Course Objective: This course introduces the basic concepts of engineering materials with an emphasis on conventional as well as smart materials required in the field of robotics. The course will describe the change in properties of smart materials in response to environmental stimuli. Also, it will discuss that the characteristics of smart materials exceed the abilities of traditional materials, especially in environments where conditions are constantly changing. This course is designed to provide an integrated and complete knowledge of materials used in the field of robotics for different applications.				
Syllabus Basics of Material Science: Fundamental terms in materials engineering and their significance, various properties and their evaluation, categories of engineering materials (alloys, composites, ceramics etc.). Physical & Mechanical Metallurgy of Materials: Stress-strain behaviour, structure-processing-properties correlation, strengthening mechanisms, thermo-mechanical processing of materials. Conventional and Advanced Materials: Properties and applications of conventional materials like alloy steels, thermoplastic fibers, and metal matrix composites. Materials for soft robotics/soft actuators viz. elastomers, shape memory polymers, polymer matrix composites. Properties and applications of smart materials viz. shape memory alloys (SMAs), high entropy alloys (HEAs), piezoelectric materials etc. Materials Selection for Specific Robot Applications: Materials selection for different parts of a robot for specific applications like farming, biomedical field, mining, materials handling, welding etc.				
Laboratory Work (if applicable) Students shall perform experiments based on - To determine the composition of a given specimen using the spectrometer. - To shot blast a given specimen to determine the change in microstructure. - To determine the grain size of given specimens using the line intercept method. - To determine the change in surface roughness of the shot blasted sample compared to the reference sample. - To determine the effect of grain size on the hardness property of given specimens. - To determine the effect of grain size on the tensile properties of given specimens. - To determine the effect of time and temperature on the geometry of shape memory alloys (SMAs). - To study the effect of electrical signal on the mechanical actuation of a piezoelectric material.				
Course Learning Objectives (CLO) The students will be able to: select the appropriate category of engineering material for a given application				

Approved in 113th meeting of the Senate held on September 7, 2024, respectively. Further, revised in 1st meeting of the Academic Council held on September 11, 2025.

2.	analyze the structure-property correlation for a particular thermo-mechanical processing route
3.	analyze the characteristics of conventional and smart materials for processing of robotic components
4.	select appropriate materials for specific robotic applications
Text Books	
1.	M. Shahinpoor, Fundamentals of Smart Materials, Royal Society of Chemistry, 2020.
2.	D. J. Leo, Engineering Analysis of Smart Material Systems, John Wiley & Sons, Inc., 2007
Reference Books	
1.	R. Smith, Smart Material Systems: Model Development, SIAM, 2005
2.	ASME Smart Materials and Adaptive Structures and Intelligent Systems proceedings 2009-2020
3.	K. Otsuka and C. M. Wayman, Shape Memory Materials, Cambridge Press 1999
4.	M Shahinpoor and Hans-Jorg Schneider, Intelligent Materials, Royal Society of Chemistry Press 2007
5.	Culshaw, B., ‘Smart Structures and Materials’ Artech House, Inc., Norwood, USA, 1996

Evaluation Scheme:

S.No.	Evaluation Elements	Weightage (%)
1.	MST	25
2.	EST	40
3.	Sessionals (May include Assignments/Projects/Tutorials/Quizzes/Lab Evaluations)	35

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